Nonintegrality of Schmidt-type coefficients

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Abstract

We prove Guo and Zeng's nonintegrality conjecture for every pair of integers $r > s > 1$. The defining equation at $n = p-2$, reduced at a prime $p \geq 5$ not dividing $r-s$, gives the obstruction. In fact the least common multiple of the first $p-1$ coefficient denominators is divisible by p^{s-1}. Thus infinitely many coefficients are nonintegral, and all but finitely many primes occur in their denominators.

Task 1: the original conjecture

For positive integers r, s, let $c_k = c_k^{(r,s)}$ be the unique rational numbers defined by

$$\sum_{k=0}^{n} B_{n,k}^{r} = \sum_{k=0}^{n} B_{n,k}^{s} c_k \quad (n \geq 0), \qquad B_{n,k} = \binom{n}{k}\binom{n+k}{k}. \tag{1}$$

The diagonal is nonzero, so this is a triangular definition. Conjecture 4.2 of Guo and Zeng [1], recorded as MathDB 323297, states that for every $r > s > 1$ there exists a positive index k for which c_k is not an integer. We prove the following stronger assertion.

Theorem 1. *Fix integers $r > s > 1$ and put $d = r - s$. For each prime $p \geq 5$ with $p \nmid d$, let D_{p-2} be the least common multiple of the positive reduced denominators of $c_0, \ldots, c_{p-2}$. Then*

$$p^{s-1} \mid D_{p-2}.$$

In particular some c_k, with $2 \leq k \leq p-2$, has reduced denominator divisible by p^{s-1}.

Proof. The cases $n = 0, 1$ of (1) give

$$c_0 = 1, \qquad c_1 = 2^{r-s} = 2^d. \tag{2}$$

Set $n = p-2$ and $a = B_{p-2,1} = (p-2)(p-1) = 2 - 3p + p^2$. For every $2 \leq k \leq p-2$, the binomial coefficient

$$\binom{p-2+k}{k} = \frac{(p-1)p(p+1)\cdots(p-2+k)}{k!}$$

contains a factor p in the numerator and none in the denominator. Consequently $p \mid B_{p-2,k}$. (The numerator has exactly k factors; the displayed product begins at $p-1$.)

Write $D = D_{p-2}$. After using (2), multiplying the equation at $n = p-2$ by D, and separating $k = 0, 1$, we obtain the integer identity

$$Da^s(a^d - 2^d) + D\sum_{k=2}^{p-2} B_{p-2,k}^{r} = \sum_{k=2}^{p-2} B_{p-2,k}^{s}(Dc_k). \tag{3}$$

Every Dc_k is an integer. The right side is divisible by p^s. The second term on the left is divisible by p^r, and $r > s$. It follows that

$$p^s \mid Da^s(a^d - 2^d). \tag{4}$$

Since $a \equiv 2 \pmod{p}$, $p \nmid a$. The binomial theorem gives

$$a^d - 2^d \equiv (2 - 3p)^d - 2^d \equiv -3d\,2^{d-1}p \pmod{p^2}.$$

The coefficient $-3d2^{d-1}$ is nonzero modulo p, because $p \geq 5$ and $p \nmid d$. Thus $a^d - 2^d$ is divisible by p but not by p^2. The divisibility (4) therefore forces $p^{s-1} \mid D$. An lcm attains each prime exponent at one of its factors, which proves the last assertion. The indices 0 and 1 have denominator 1 by (2). □

Corollary 1. *Guo and Zeng's Conjecture 4.2 holds. More strongly, for each fixed pair $r > s > 1$ there are infinitely many indices k with $c_k^{(r,s)} \notin \mathbb{Z}$. Every prime $p \geq 5$ not dividing $r - s$ divides the reduced denominator of some coefficient, to at least the power $s - 1$.*

Proof. There are infinitely many primes not dividing the fixed nonzero integer $6(r - s)$. Apply Theorem 1 to each. A finite collection of rational numbers has only finitely many prime factors in its reduced denominators. Hence infinitely many distinct nonintegral coefficients are necessary. In particular at least one exists, as conjectured. □

An explicit upper bound for the first nonintegral index is $p_0 - 2$, where p_0 is the least prime at least 5 not dividing $r - s$. Also, writing D_N for the denominator lcm through index N, we immediately obtain

$$\prod_{\substack{5 \leq p \leq N+2 \\ p \text{ prime},\ p \nmid (r-s)}} p^{s-1} \;\mid\; D_N.$$

These claims follow from the same proof and receive no separate problem credit.

Scope, source and checks

The full current author manuscript and the full final published paper were read; the definition and Conjecture 4.2 agree. The different finite integrality conjecture, Conjecture 4.1, has its own proof package and is not counted again here. The theorem above is a direct proof for all integers $r > s > 1$; it does not depend on that other result or on computation.

The accompanying `verify_denominators.py` performs supplementary exact checks of the binomial divisibility, first-order obstruction, and actual rational coefficient denominators over stated finite ranges. These checks test conventions and edge cases; the universal conclusion is proved above. The literature audit records a dated, bounded search for prior claims. No independent peer review or formal proof-assistant verification is claimed.

Four further research tasks

Task 2. For each fixed $r > s > 1$, determine the exact largest p-power in D_{p-2} for every $p \geq 5$, including primes dividing $r - s$.

Task 3. Determine the asymptotic growth of $\log D_N$ for each fixed pair $r > s > 1$, sharpening the prime-product lower bound above.

Task 4. Use the equations at $n = p - m$, with fixed $m \geq 3$, to derive additional necessary congruences on r, s for integrality through $p - m$, and compare their strength with the $m = 2$ obstruction.

Task 5. Determine the first nonintegral index as a function of (r, s) and characterize the pairs for which the bound $p_0 - 2$ is attained.

References

[1] V. J. W. Guo and J. Zeng, *On Zudilin's q-question about Schmidt's problem*, Electronic Journal of Combinatorics **19**(3) (2012), P4, doi:10.37236/2231. Conjecture 4.2 appears on page 4 of the final article; current author manuscript: arXiv:1204.0187v3.